

Checkpoint 2: DataLoader and Diffusion Pipeline

Objective

Show that paired images enter the model correctly and that the diffusion training task is implemented.

Key Message

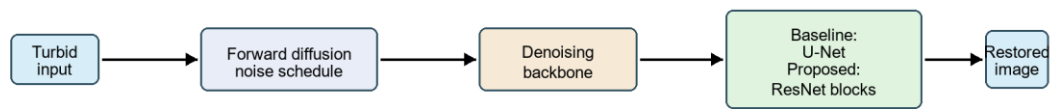
The data pipeline, noising process, and tests are ready before full model training.

Completed Work

- Implemented normalized paired image loading.
- Implemented timestep sampling and linear beta schedule.
- Added sanity tests for dataset pairing, diffusion, and training steps.

Evidence

Component	Status
Paired DataLoader	Completed
Forward diffusion	Completed
Unit tests	Passing



Same paired dataset and evaluation protocol for both denoising backbones

Conditional diffusion restoration pipeline.

Notes

- The same diffusion objective is used for both baseline and residual models.